

Rocket data checklist

Fill in the value column. Lengths in mm from the nose tip, masses in g, angles in deg, times in s or ms. Attach raw logs for the thrust curves and the igniter tests.

Measure first (these decide the landing)

Item	Unit	Accuracy	Value
☐ Landing motor thrust curve, 2+ motors, load cell log	N vs s	1-2 %	
☐ Landing motor propellant mass / total mass	g / g	1 g	
☐ Igniter delay: mean and spread (about 20 igniters)	ms / ms	10 ms	
☐ Rocket mass without motors, each flight day	g	10 g	
☐ Brake petals drag area (plate area x 1.2, or drop test)	cm2	5 %	

Airframe

Item	Unit	Accuracy	Value
☐ Dry mass (no motors)	g	10 g	
☐ Centre of gravity from nose tip, no motors	mm	5 mm	
☐ Pitch inertia about that point	kg m2	10 %	
☐ Roll inertia about the body axis	kg m2	20 %	
☐ Length, nose tip to tail end	mm	2 mm	
☐ Body diameter	mm	0.5 mm	

Aerodynamics (OpenRocket)

Item	Unit	Accuracy	Value
☐ Drag coefficient, petals shut	-	20 %	
☐ Normal force slope	per rad	20 %	
☐ Centre of pressure from nose tip	mm	10 mm	

Motors

Item	Unit	Accuracy	Value
☐ Ascent motor .eng file or thrust curve	N vs s	5 %	
☐ Ascent igniter delay: mean and spread	ms / ms	20 ms	
☐ Ascent motor centre of mass from nose tip, installed	mm	5 mm	
☐ Landing motor centre of mass from nose tip, installed	mm	5 mm	
☐ Which motors sit in the gimbal mount	yes / no	-	

Gimbal and servos

Item	Unit	Accuracy	Value
☐ Gimbal pivot from nose tip	mm	5 mm	
☐ Maximum gimbal angle, both axes	deg	0.5 deg	
☐ Servo speed	deg/s	20 %	
☐ Servo delay, command to motion	ms	5 ms	
☐ Servo deadband	deg	0.1 deg	
☐ Pulse width at gimbal centre, pitch servo / yaw servo	us / us	5 us	
☐ Pulse change per servo degree	us/deg	5 %	
☐ Direction sign, pitch / yaw (longer pulse leans nose to +x / +y?)	+1 or -1	-	
☐ Linkage ratio: gimbal degrees per servo degree	-	5 %	
☐ Pulse min / max / resolution	us	exact	

Drag brake

Item	Unit	Accuracy	Value
☐ Petal station from nose tip (must be ahead of the burnout CG)	mm	5 mm	
☐ Time to open fully / to shut	s / s	0.1 s	

Landing legs

Item	Unit	Accuracy	Value
☐ Foot circle diameter	mm	5 mm	
☐ Feet below the tail end	mm	5 mm	
☐ Number of legs	-	-	
☐ Touchdown speed the legs survive, down / sideways (drop test)	m/s / m/s	0.2 m/s	
☐ Tilt the legs survive	deg	1 deg	

Sensors (30 s log with the board still on a table)

Item	Unit	Accuracy	Value
☐ IMU sample rate / lag	Hz / ms	exact / 1 ms	
☐ Accelerometer noise / bias / range	m/s ²	20 %	
☐ Gyro noise / bias / range	deg/s	20 %	
☐ Barometer rate / lag / noise / bias	Hz, ms, m, m	20 %	

Launch site

Item	Unit	Accuracy	Value
☐ Site elevation	m	20 m	
☐ Wind speed and direction on the day	m/s, deg	0.5 m/s	